

A Journey through Local Class Field Theory

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Preface

This report is based on everything I learnt under the guidance of Prof. Sandeep Varma as a part of the VSRP program conducted by TIFR. The results mentioned here can be found either in the book Local Fields by Jean-Pierre Serre([\[Serre\]](#)) or Algebraic Number Theory by J.W.S. Cassels and A. Fröhlich([\[Cassels\]](#)). For a more detailed treatment of the formulas of Homological Algebra, one can refer to Homological Algebra by Henri Cartan and Samuel Eilenberg([\[Cartan-Eilenberg\]](#)).

I would like to thank Prof. Varma for giving me this incredible opportunity to work under him and am immensely grateful for his guidance and for the amount of time he spent with me explaining different concepts. Through sessions spanning multiple hours, he generously devoted his time to teach and mentor us. Additionally, his availability and responsiveness through email allowed for continuous and fruitful exchanges. This project has sparked my interest in the Class Field Theory and its interaction with other areas in Number Theory, and am motivated to explore this topic further in my future studies due to the profound influence of his guidance. I would also like to thank the organizers of VSRP for this wonderful program and for giving me the opportunity to take part in VSRP-2025. I would also like to thank Ruchira Mukherjee for her contributions and discussions.

Undertaking this project has been a challenging and enriching experience for me. I eagerly intend to pursue related topics in my future research endeavors.

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Local Class Field Theory

Let G be a group, written multiplicatively, and let A be an abelian group, written additively. We would be concerned with the study of G -modules or equivalently $\mathbb{Z}[G]$ modules. Let A be a G -module, and let A^G be the submodule consisting of the elements fixed by G . If $f : A \rightarrow B$ is a G -homomorphism, then f maps $f : A^G \rightarrow B^G$ hence we can speak of the functor A^G which is an additive functor that is left exact. By definition, the right derived functors of the functor A^G are the cohomology groups of G with coefficients in A ; they are denoted $H^n(G, A)$, $n \geq 0$. $H^n(G, A)$ are not just sequence of functors but a cohomological functor.

In particular we are interested in three maps namely Restriction (Res), Inflation (Inf) and Corestriction (Cor).

- Res: $H^n(G, A) \rightarrow H^n(H, A)$ where H is a subgroup of G
- Inf: $H^n(G/H, A^H) \rightarrow H^n(G, A)$ where H is a normal subgroup of G
- Cor: $H^n(H, A) \rightarrow H^n(G, A)$ where H is a finite index subgroup of G

If $a \in A^H$ and $s \in G$, the element sa depends only on the left coset of $s \bmod H$; we take G/H to be finite, and hence we can form the sum

$$N_{G/H}(a) = \sum_{s \in G/H} sa \quad (1)$$

which we call as norm of a .

Let G be a finite group. In the group algebra $\mathbb{Z}[G]$, the element $\sum_{g \in G} s$ will be called the norm denoted as N . For every G -module norm defines an endomorphism from A to A by the formula

$$N(a) = \sum_{s \in G} sa. \quad (2)$$

We also define I_G to be the augmentation ideal of $\mathbb{Z}[G]$, i.e the linear combination of $s - 1$, $s \in G$.

Now we set to define the Tate cohomology groups of G as follows -

$$\textbf{Definition 1. } \hat{H}^n(G, A) = \begin{cases} H^n(G, A), & \text{for all } n \geq 1 \\ \frac{A^G}{N(A)}, & n = 0 \\ \frac{Ker N_A}{I_G A}, & n = -1 \\ H_n(G, A), & \text{for all } n \leq -2 \end{cases}$$

where $Ker N_A$ is the kernel of the norm map from A to itself. $H_n(G, A)$ is the n^{th} homology group of G defined as left derived functor of the additive right exact functor $A_G = \frac{A}{DA}$ where DA be the subgroup of A generated by the $s.a - a$, $s \in G$ and $a \in A$.

Remark 1. When G is finite cyclic group then the groups $\hat{H}^n(G, A)$ depends only on the parity of n .

Definition 2. Let G be a finite group and A a G -module. A is called cohomologically trivial if, for every subgroup H of G and every $n \in \mathbb{Z}$, $\hat{H}^n(H, A) = 0$.

We then have a very interesting result which we state next as a proposition -

Proposition 1. (*T.Nakayama*) Let G be a finite group and A be any G -module. A is cohomologically trivial if and only if it has projective dimension at most 2.

This is a very beautiful result in itself. The study of group cohomology which looked very distant at first is actually being used to prove some results about modules over the ring $\mathbb{Z}[G]$. The proof of the theorem is very enlightening and intriguing. For details please refer [Serre] or [Cassels] or [Nakayama] and [T.Nakayama].

Then we looked at a very important result of Tate Nakayama [Serre]. We state only a corollary of this beautiful result which we state as a theorem and that we need during class formations.

Theorem 1. ([Tate]). For every integer n and every subgroup H of G , the cup product defined with $a_H = \text{Res}(a)$ defines an isomorphism

$$\hat{H}^n(H, \mathbb{Z}) \rightarrow \hat{H}^{n+2}(H, A). \quad (3)$$

Next we move to the notion of formation for a more detailed treatment one can look up [Serre] or [Artin-Tate].

We begin with a group G and a family of subgroups $\{G_E\}_{E \in X}$, of finite index in G , we assume the indexing to be such that $G_E = G_{E'}$ implies $E = E'$. We further assume that -

1. For every finite family F_i of elements of X , there exists $F \in X$ such that $G_F = \bigcap G_{F_i}$.
2. Every subgroup G' of G that contains a subgroup G_F , $F \in X$ has the form $G' = G'_{F'}$ for some $F' \in X$.
3. If $s \in G$ and $F \in X$, the subgroup $s.G.s^{-1}$ has the form $G_{F''}$ for some $F'' \in X$.

For G and $\{G_E\}_{E \in X}$ satisfying the above conditions, we will transfer to X the terminology of Galois theory. The elements of X will be called “fields”. We write $E \subset F$ if $G_F \subset G_E$, we say “ F/E is a Galois extension” if G_F is normal in G_E , and we call the quotient group G_E/G_F “the Galois group $G(F/E)$ ”. Next we suppose let A can be considered as a topological G -module, i.e for $a \in A$ the stabilizer of a is one of the subgroups G_E , $E \in X$. A formation is then a system $(G, \{G_E\}_{E \in X}, A)$ satisfying the above conditions. For $E \in X$ A_E denotes the subgroup of A fixed by the group G_E .

As the group $G(F/E)$ acts on A_F the cohomology groups $H^n(G(F/E), A_F)$ is defined and we denote it by $H^n(G(F/E))$. Similarly the homology and Tate cohomology groups are denoted as $H_n(G(F/E))$ and $\hat{H}^n(G(F/E))$ respectively.

Definition 3. A class formation consists of a formation $(G, \{G_E\}_{E \in X}, A)$ together with a homomorphism

$$\text{inv}_E : H^2(F/E) \rightarrow \mathbb{Q}/\mathbb{Z} \quad (4)$$

for each Galois extension F/E satisfying the following axioms -

1. For every Galois extension F/E , $H^1(F/E) = 0$.
2. The homomorphism inv_E is injective, and maps $H^2(F/E)$ onto the unique subgroup of $\mathbb{Q}/\mathbb{Z}$ of order equal to $[F:E]$.
3. For any extension E'/E we have
 $inv'_E \circ Res_{E'/E} = [E' : E].inv_E$.

If F/E is any Galois extension, and $n = [F : E]$ is its degree, then there exists a unique element $u_{F/E} \in H^2(F/E)$ such that $inv_E(u_{F/E}) = 1/n$. It is called the fundamental class of the extension F/E . Now from [Theorem 1] we get the following,

Theorem 2. *For every Galois extension F/E and every integer n , the homomorphism*

$$\theta^n(F/E) : \hat{H}^n(H, \mathbb{Z}) \rightarrow \hat{H}^{n+2}(H, A) \quad (5)$$

defined by $x \rightarrow u_{F/E} \cup x$ (cup product) is bijective.

We are interested in $n=-2$, where θ^{-2} gives us an isomorphism between

$$G_{F/E}^{ab} \rightarrow A_E/N_{F/E}A_F. \quad (6)$$

The inverse of the isomorphism defined in [Theorem 2] is called the local reciprocity isomorphism or norm residue symbol and denoted as

$a \mapsto (a, F/E)$ where $a \in A_E$ and $(a, F/E) \in G_{F/E}^{ab}$. [Cassels]

Definition 4. A subgroup I of A_E is called a norm group if there exists an extension E'/E such that $I = N_{E'/E}(A'_E)$.

For ease of notation if F/E is abelian we denote $N_{F/E}A_F$ by I_F . We now present a proposition that would be useful later.

Proposition 2. *The map $F \rightarrow I_F$ is a bijection of the set of abelian extensions of E onto the set of norm groups of A_E ; this correspondence reverses inclusion and satisfies*

$$I_{F.F'} = I_F \cap I_{F'} \text{ and } I_{F \cap F'} = I_F + I_{F'}.$$

Furthermore, every subgroup of A_E which contains a norm group is itself a norm group.

Now we give one the most important theorem of Local Class Field Theory the Existence Theorem. To this end, we assume given a topology on each A_E compatible with its group structure, and such that if $E \subset F$, then the topology on A_E is induced from that of A_F : we also assume that for every $s \in G$ the map $s \mapsto s.a$ is a continuous map of A_E into A_{sE} . These assumptions imply that the norm map $N_{F/E} : A_F \rightarrow A_E$ is continuous.

Definition 5. The universal norm group of E (denoted by D_E) is the intersection of all the norm groups of A_E .

We now are going to impose three axioms on these topologies, and we will show that the norm groups are precisely the closed subgroups of finite index in A_E which we state in [Theorem 3].

Axioms:

1. For any extension F/E , the norm homomorphism
 $N_{F/E} : A_F \rightarrow A_E$
has closed image and compact kernel.
2. For every prime number p , there exists a field E_p such that, for $E_p \subset E$, the map
 $\varphi_p : x \mapsto px$ of A_E into itself satisfies the condition that The kernel of φ_p is compact
and its image contains D_E .
3. There exists a compact subgroup U_E of A_E such that every closed subgroup of finite
index in A_E that contains U_E is a norm group.

Theorem 3. (*Existence Theorem*) Suppose the above axioms hold. For a subgroup of A_E to be a norm group, it is necessary and sufficient that it be closed and of finite index.

Now finally we put ourselves in the context of Local Class Field theory. K denotes a field that is complete under a discrete valuation v and having quasi-finite residue field $\bar{K}$. First choose a separable closure K_s of K , and let X be the set of finite subextensions of K_s . Let $G = G(K_s/K)$ and, for each $E \in X$ let $G_E = G(K_s/E)$ be the subgroup corresponding to E . The group G acts on K_s^* ; and we obtain thereby a formation. To check its a class formation we need to check the axioms of [Definition 3], which are easy to check the first axiom just being Hilbert's Theorem 90 !!!

Now to prove Existence Theorem ([Theorem 3]) in context of Local Class Field Theory we just need to check the axioms stated after [Definition 5], which are not hard to check. Thus we have proved

Theorem 4. (*Existence Theorem for Local Class Field Theory*) If K be a local field then every closed subgroup of finite index in K^* is a norm group.

(In other words, if H is such a subgroup, there exists a finite abelian extension L/K such that $N_{L/K}(L^*) = H$; moreover, if such an extension exists, we know that it is unique ([Proposition 2]).

Using [Theorem 4] to $\mathbb{Q}_p$ (the completion of $\mathbb{Q}$ with respect to p -adic metric and its residue field being $\mathbb{Z}/p\mathbb{Z}$ which is fine and hence $\mathbb{Q}_p$ is a local field), we derive a very important theorem which we state next.

Theorem 5. Let p be prime, and let E be the field obtained by adjoining all the roots of unity to $\mathbb{Q}_p$. Then E is the maximal abelian extension of $\mathbb{Q}_p$.

This is what is the celebrated Local Kronecker–Weber theorem. We use [Theorem 5] to prove another more common version of Local Kronecker–Weber theorem which we state next.

Theorem 6. (*Kronecker–Weber*) Every finite abelian extension of $\mathbb{Q}_p$ is contained in a cyclotomic extension of $\mathbb{Q}_p$.

Proof. We know

$$\mathbb{Q}_p^{ab} = \bigcup_{L/\mathbb{Q}_p \text{ finite abelian extension}} L. \quad (7)$$

where $L \subset \mathbb{Q}_p^{sep}$. Now given the description of $\mathbb{Q}_p^{ab}$ from [Theorem 5] we get

$$\mathbb{Q}_p^{ab} = \bigcup_{n \geq 1} \mathbb{Q}_p(\zeta_n) \quad (8)$$

where ζ_n is the n th root of unity. Now let $L/\mathbb{Q}_p$ be a finite abelian extension of $\mathbb{Q}_p$. By primitive element theorem $L = \mathbb{Q}_p(\alpha)$ for some $\alpha \in L$. Now since $L \subset \mathbb{Q}_p^{sep}$, we have $\alpha \in \mathbb{Q}_p^{ab}$. But if $\alpha \notin \mathbb{Q}_p(\zeta_n)$ for all $n \geq 1$ then by equation (8) we get $\alpha \notin \mathbb{Q}_p^{ab}$, which is a contradiction. Hence we must have some $n \geq 1$ such that $\alpha \in \mathbb{Q}_p(\zeta_n)$ and consequently we have $\mathbb{Q}_p(\alpha) \in \mathbb{Q}_p(\zeta_n)$. $\square$

Retrospect:

Now that we have taken a brief journey through Local Class Field Theory, it's worthwhile to pause and reflect

-from where did we come, and where are we headed?

We began by exploring the structure and properties of the group ring $\mathbb{Z}[G]$, which naturally led us to the study of group cohomology. Building upon that, we delved into the Tate cohomology groups, which emerged as a central tool in the development of Local Class Field Theory. Along the way, we encountered the celebrated Tate–Nakayama theorem—a pivotal result that bridges cohomological techniques with arithmetic applications.

We then made a brief excursion into Galois cohomology, and introduced the powerful framework of Class Formations—an elegant way to organize the structures needed for class field theory.

While time didn't allow us to explore the Brauer group and its deep connections to Division algebras, armed with the notion of Class Formation, we proceeded to prove the Existence Theorem of Local Class Field Theory. And finally, we ended on a classical note, with the Local Kronecker–Weber theorem, bringing centuries of arithmetic into a unifying perspective.

Of course, the story doesn't end here. What lies ahead is a vast and rich landscape: Global Class Field Theory, deeper aspects of Galois cohomology, and its remarkable connections with elliptic curves, modular forms, and beyond. These are topics I aspire to study in the days to come.

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